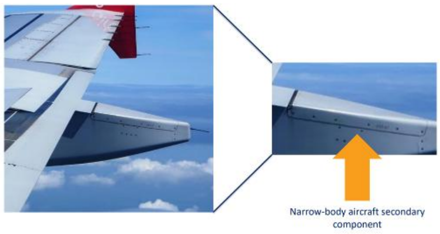
CASE STUDY 2



PROBLEM

Great at service but issues with end of life and residual waste

GOAL

Produce an Aircraft Flap Track Fairing

Aircraft Flap Track Fairing =

- Secondary component
- Allows the flap track to extend and contract

DESIGN REQURIEMENTS

- Lightweight to reduce aircraft weight and fuel consumption
- High strength and stiffness to resist aerodynamic forces
- Durability to withstand environmental conditions at 42,000 ft
- Manufacturability with minimal waste and cost efficiency
- Sustainability: Low environmental impact and recyclability
- ‘L’ shape structure -be able to produce numerous exact ‘L’ shape components

THEIR SOLUTION

- M21E aero grade cf pre-preg specifically manufactured to meet the strict requirements for use on aircraft bodies – including, performance, safety and reg requirements (qualified materials)
- Epoxy
- Foam core: rPET
- Sandwich panel structure with ply orientated lay up
- Tool ‘on site’ at GKN (hand lay up for this secondary structure)
- Debulk with consumables
- Autoclave cure (more than one)
- Trim excess – overlay

POSSIBLE QUESTIONS

What does the fairing do in service?

- It reduces drag, protects the underlying systems within the wing of the aircraft from environmental exposure and structural damage.
- Optimize fuel efficiency while increases the aircraft component longevity
- At take off and landing the flap movements generates turbulent flows and localized pressure differentials

- **Aerodynamic Efficiency**: It reduces parasitic drag by smoothing out the junction between the wing and the body.
- **Component Protection**: Shields internal components such as the flap track mechanism from environmental exposure like rain, dust, and temperature fluctuations.
- **Structural Role**: Although often non-load-bearing, it can provide minor structural support or stiffness.
- **Aesthetic Integration**: Enhances visual continuity between aircraft components.

How does the L section of the fairing connect to the wing?

- Mechanically fastened using bolts or rivets along structural ribs or spars.
- Bonded using structural adhesives to ensure tight integration while minimizing stress concentrations.
- Designed with overlap or step joints to allow for efficient load transfer and avoid sharp aerodynamic transitions.

How you choose the materials for the fairing – what are we looking to optimise?

- Low weight to reduce fuel consumption and emissions.
- High stiffness to maintain shape and resist aerodynamic and structural loads.
- Thermal and chemical resistance for durability across climates and altitudes.
- Sustainability, increasingly critical in design—considering recyclability and environmental impact.
- Regulatory compliance, using aerospace-certified materials (e.g., M21E pre-preg carbon fiber)

Material index: (if modelled as a beam, but could be modelled as a panel and would be slightly different)

- Minimise mass
- Maximise stiffness and strength

Minimise:

$$M_1 = \frac{\rho}{E^{\frac{1}{2}}} \quad (Eq\ 1) \qquad\qquad M_2 = \frac{\rho}{\sigma_Y^{\frac{2}{3}}} \quad (Eq\ 2)$$

What are the requirements?

- Light and stiff in bending
- Protect the flap track mechanisms
- Provide aerodynamic cover to reduce drag
- Resist conditions at cruising altitude and on the ground in all operating conditions globally

- Mass limit: Total component should not exceed 3.40 kg; material mass should be ~3.02 kg.
- Deflection constraint: Max deflection at load = 1.66 mm.
- Environmental resilience: Must perform in varied global climates at altitude and on ground.
- Durability and safety compliance: Must meet strict aerospace regulations and testing

How can we increase the flexural stiffness whilst keeping the mass low?

Flexural stiffness EIEIEI can be increased by:

- Increasing the second moment of area (I):
    - Use sandwich panels: a lightweight core (like recycled PET foam) with stiff composite skins significantly increases I.
    - Adjust geometry (e.g., L-shaped section, corrugation) for structural efficiency.
- Using high E materials:
    - Carbon fiber composites (high modulus fibers).
- Optimizing ply orientation: Layups at 0°, ±45°, and 90° increase stiffness in multiple directions

- Introduce a sandwich structure, an inner foam core and outer stiffer and stronger panel.
- Inner core = Recycled PET Foam, lightweight to simply increase the distance between the panels to enlarge the second moment of area
- Outer skins= M21E Carbon fibre composite, aerospace grade of pre-impregnated (pre-preg) to meet aircraft requirements

Deflection Calculation (beam analogy): $\delta = (P \times L^3) / (48 \times E \times I)$

Where:

- δ = max deflection ($\leq$ 1.66 mm)

- P = applied load

- L = length of fairing (~1.2 m)

- E = effective modulus of laminate

- I = second moment of area (improved using a sandwich structure)

Which materials are best for this application?

M21E Carbon Fiber Prepreg:

- Aerospace-grade, epoxy-matrix, unidirectional carbon fibers.
- Provides excellent stiffness-to-weight ratio and durability

Core: Recycled PET (rPET) foam:

- Lightweight, sustainable, good compressive properties.

Breakdown of materials:

- M21E = approved aerospace grade of pre-impregnated (pre-preg)
  - Pre-preg= carbon fibred that are pre impregnated with resin before it is shipped out
  - Resin system includes curing agent making it suitable to be readily laid in the mould without additional resin
  - High strength to weight ratio
  - Lighter than metals such as aluminium
  - Low thermal expansion suitable for fluctuation conditions
- Matrix = epoxy resin,
  - polymers that have excellent compressive and shear properties
  - Easy to work with, cut, bend, shape to desired pattern
  - Toughened epoxy matrix, more suitable under thermal cycling and fatigue
- Unidirectional Carbon fibres= non-woven,
  - Fibres lay parallel to one another, no potential for gaps to form unlike in woven
  - has superior strength compared to woven fibres, where the fibres run in two directions. In woven fibres, crimping occurs which causes the fibres to bend under stress force which reduces stiffness and strength
  - Using a UD lay-up means that the fibres can lay flat, reducing risks of air pockets
- Recycled PET (polyethylene terephthalate) foam core =
  - High compression strength
  - High shear strength
  - Thermoplastic
  - High strength to weight ratio

How you make them?

Prepreg Material Handling and Storage:

Prepreg (pre-impregnated composite fibers) must be stored in a freezer to slow the curing reaction - eady-to-use materials, combining reinforcement (carbon fiber) and resin in precise ratios — but they are time-sensitive:

Once removed:

  - It has a limited out-time (typically 5 days) at room temperature before it begins to cure and must be scrapped.
  - This means the layup process must be completed quickly and efficiently to avoid material waste.

  - The prepregs are taken out of the freezer just before layup, thawed under controlled conditions, and then laid by hand.

Prepregs also have a finite freezer shelf life so stock rotation and usage planning are essential.

Autoclaves are not run for just one part: To conserve energy and reduce costs, parts are batched together for curing.

How to make individual panels for a Sandwich panel structure with ply-oriented layup:

1.Hand lay up

Each skin is constructed using a ply-oriented layup (m **three carbon fiber plies**, laid in specific orientations):

- Three fiber orientations per skin: 0°, 90°, and ±45°
  - 0° fibers: aligned with the primary load path → maximum stiffness and strength in that direction.
  - 90° fibers: allow resistance to loads perpendicular to the main direction → aids flexural performance.
  - ±45° fibers: form a quasi-isotropic layer when combined with 0° and 90°, offering balanced properties in all directions.
    - This improves impact tolerance, damage resistance, and reduces delamination risk.
    - Ensures consistent mechanical performance regardless of load direction.

The composite fabric is intentionally laid **beyond the final trim line**, wrapping slightly **over the foam core edges**. This ensures **full coverage**, helps avoid edge delamination, and allows for a clean, accurate final shape after trimming.

2. Vacuum Bagging

“Each layer is degassed by vacuum before the next layer (ply) is applied”

- Air and volatile gases trapped between layers are removed using a vacuum process.
- Vacuum bagging:
  - Flexible vacuum bags are placed over the layup.
  - Sealed and pressurized to evenly compact fibers and eliminate voids.
  - Promotes excellent fiber-resin adhesion and dimensional stability.

Result: denser, stronger laminate with reduced porosity → essential for aerospace applications . Ensure **full resin contact and compaction**. Prevent voids or resin-rich zones that weaken the part.

3. Foam Core Integration (rPET)

A lightweight recycled PET foam is inserted between the composite skins:

- Provides low density while significantly increasing the second moment of area, improving bending stiffness.
- Sustainable material choice that supports circular manufacturing practices.

4. GKN Tool for Layup

A specialized mold/tool provided by GKN is used to ensure correct shape and precision.

Facilitates repeatable and accurate hand layup, critical for maintaining quality and aerodynamic profile.

5. Autoclave Curing

- The entire layup is cured in an autoclave with internal vacuum:
  - Heat and pressure are applied simultaneously.
  - Promotes full polymer crosslinking of the epoxy resin.
  - Results in high fiber volume fraction and minimal voids.
- This step is crucial to meet:
  - Structural requirements under service loads.
  - Environmental durability (e.g., withstand forces at 42,000 ft altitude).

6. Trimming and Finishing

- The layup includes an overlay, or margin beyond final dimensions, to account for edge imperfections.
- After curing, the excess is trimmed off precisely to final specs using CNC or manual methods.
- This ensures aerodynamic fit, uniform edges, and assembly compatibility.

How might you test them?

Non-Destructive Testing (NDT) - Ultrasound / X-ray / Thermography

Inspect inside the fairing for:

  - Air bubbles (voids)
  - Delamination
  - Incomplete curing

Done before and after mechanical tests.

Wind Tunnel or CFD

For aerodynamic validation if your design is new or optimized for drag: Use CFD first (cheaper, faster). If needed, test a scaled or full-size prototype in a wind tunnel.

Why Does the Problem Exist? (Composite Waste in Fairing Manufacturing)?

10%–35% of purchased prepreg material becomes un-cured offcut waste during trimming and layup.

This is due to:

- Overlay margins needed for precise trimming after curing.
- Irregular shapes and overlaps required for hand layup and ply orientation.
- Strict aerospace tolerances, which demand conservative trimming.

Minimal Value Retention of Waste

High-performance carbon fiber offcuts are expensive to produce, but once cut, their value drops dramatically.

Current recycling/reuse systems fail to retain fiber length, orientation, or resin quality — leading to low-grade reuse options or disposal.

End-of-Life Pathways Are Environmentally Damaging

Most composite offcuts are sent to:

  - Landfill, where they persist for decades.
  - Incineration, which contributes to carbon emissions and energy loss.

There is a lack of scalable infrastructure for circular use of aerospace-grade composites.

Decarbonsisng use-phase of prpoducts through alternative fuel sources will not hit Net Zero targets alone

While switching to sustainable aviation fuel helps, it doesn’t address embedded carbon in materials and manufacturing.

Decarbonizing the full life cycle, including manufacturing waste, is essential to meet Net Zero targets.

Many sustainable technologies are not cost effective even at a scale

Big changes to production systems (e.g., automated patch reuse, certification for recycled parts) are:

  - Technically complex
  - Require new tooling, validation, and qualification
  - Involve high upfront investment

What are the Areas for Decarbonisation of Aerospace (Outside of New Fuels)?

Decarbonising Carbon Fiber (CF) Production- not done in the UK , only 50% is made in UK, rest is imported.

Carbon fiber is energy-intensive to produce:

- Requires high-temperature processing (often above 1,000°C).
- Often uses fossil fuel-derived precursors like PAN (polyacrylonitrile).

Opportunities for decarbonisation:

- Switching to renewable energy for manufacturing processes.
- Using bio-based or recycled carbon precursors.
- Improving process efficiency through closed-loop energy systems.

Production and End-of-Life (EOL) Waste

- A significant portion of CF composite waste is generated during:
  - Manufacturing (10–35% material offcut).
  - End-of-life disposal (landfill or incineration).
- Strategies to reduce lifecycle emissions:
  - Design for disassembly and material recovery.
  - Implement reuse strategies (e.g., patching offcuts into secondary parts).
  - Develop certified recycling routes that retain material performance.
  - Innovate bio-degradable matrix systems or recyclable thermoplastics.

What is required to begin looking at a circular solution

Maximise Value Retention

- Reuse offcuts without damaging fiber length or orientation.
- Use precision patch systems to retain mechanical performance.

Improve Overall Sustainability

- Consider the full lifecycle: material sourcing, energy use, waste, and end-of-life.
- Aim for a net reduction in environmental impact.

Maintain or Reduce Component Mass

- Use lightweight cores (e.g., rPET) and efficient layups.
- Avoid adding extra material just to accommodate waste reuse.

Avoid Production Disruption

- Fit circular methods into current manufacturing processes.
- Ensure no delays or major retraining/tooling changes.

Ensure Cost-Effectiveness

- Must be economically viable: reduce raw material cost, scrap, and disposal fees.
- Supports long-term adoption and investment.

How can you exploit waste / recycle stream materials to make them?

The trimming process generates 10–35% waste from carbon fiber prepreg offcuts and excess overlays.

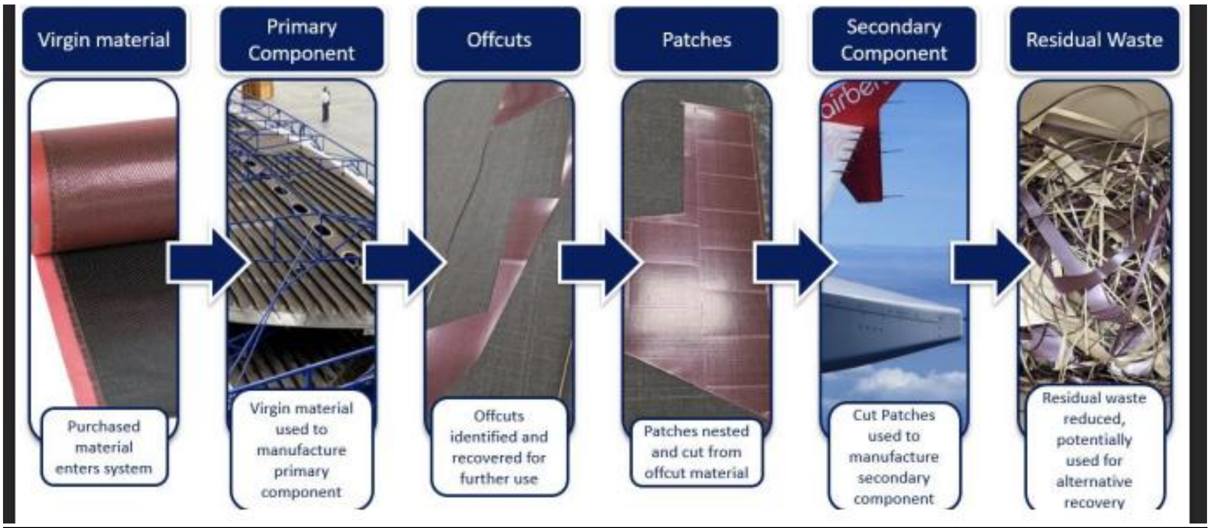
Current disposal routes often include incineration or landfill.

Sustainable alternatives:

- Patch-based reuse systems like Cevotec, which convert offcuts into smaller parts or secondary components.

Patch-Based Waste Reuse

- During primary component production, prepreg offcuts (still usable but no longer certifiable for primary use) are created.
- These offcuts can be collected, identified, and cut into precise "patches" to be reused.
- Patches are then nested and assembled into new layups — a process ideal for less critical parts like fairings.



Cevotec Project – Fibre Patch Placement Solution

A robotic arm picks up individual prepreg patches and lays them down automatically onto a tool surface.

The prepreg is not virgin material — it may be slightly older, softer, or tackier, requiring precise handling.

A software package was created to design the layup: Produces a "patchwork" structure using overlapping strips or shapes. Ensures consistent orientation and structural continuity across the part.

Why This Works

- It retains fiber length and orientation, maintaining much of the material's strength.
- Helps avoid waste of expensive carbon fiber prepreg that would otherwise go to landfill or incineration.
- Allows offcuts to be reused in secondary aerospace parts where certification thresholds are less demanding.

Challenges

- This process still requires qualification to be adopted in certified aerospace manufacturing.
- A rigorous validation and testing process is needed to prove:
    - Mechanical reliability of patchwork layups.
    - Bonding consistency.
    - Long-term durability under operational conditions.

By using robot-assisted patchwork layup of prepreg offcuts (with partners like Cevotec), waste materials can be repurposed into secondary aerospace components. This approach reduces landfill, retains material value, and supports circular economy goals — but must meet strict aerospace certification standards to be widely adopted

- Designing nesting layouts to optimize material use and reduce waste during layup.

Extra

What other materials were considered for the fairing but deemed unsuitable (notes taken from group case studies)

Material	Density (kg/m^3)	Stiffness/Strength	GWP (kg CO_2/kg)	Pros/Cons
M21E Carbon Fibre	1600	Very High	30.59	✓ Aerospace certified, strong ✗ High GWP, expensive
rPET Foam	~100	Low (core)	Low	✓ Lightweight, recycled ✗ Low stiffness
Flax Fibre	~1500	Moderate	-1.39	✓ Biodegradable ✗ Weak, absorbs

				moisture
Aluminium	2700	High	8.68	✓ Recyclable, strong ✗ Heavy
Stainless Steel	7800	High	6–10	✓ Corrosion resistant ✗ Too heavy for fairing

Why Not GFRP for a Fairing?

- Too Thick & Heavy:
  GFRP requires thicker walls (e.g. 5 cm in wind turbine blades) to achieve needed stiffness — impractical for aircraft at altitude where weight is critical.
- Not Suitable for Altitude Stress:
  Even thick GFRP shells fail under lower-stress conditions (e.g. wind turbines). Aircraft operate under much harsher loads and environmental conditions.
- Higher Maintenance:
  GFRP is more prone to micro-fractures and requires recoating or repair, increasing lifecycle costs.
- Shorter Component Life:
  CFRP (Carbon Fibre Reinforced Polymer) provides superior durability, extending the lifetime of components like fairings and reducing replacements (e.g. bearings).